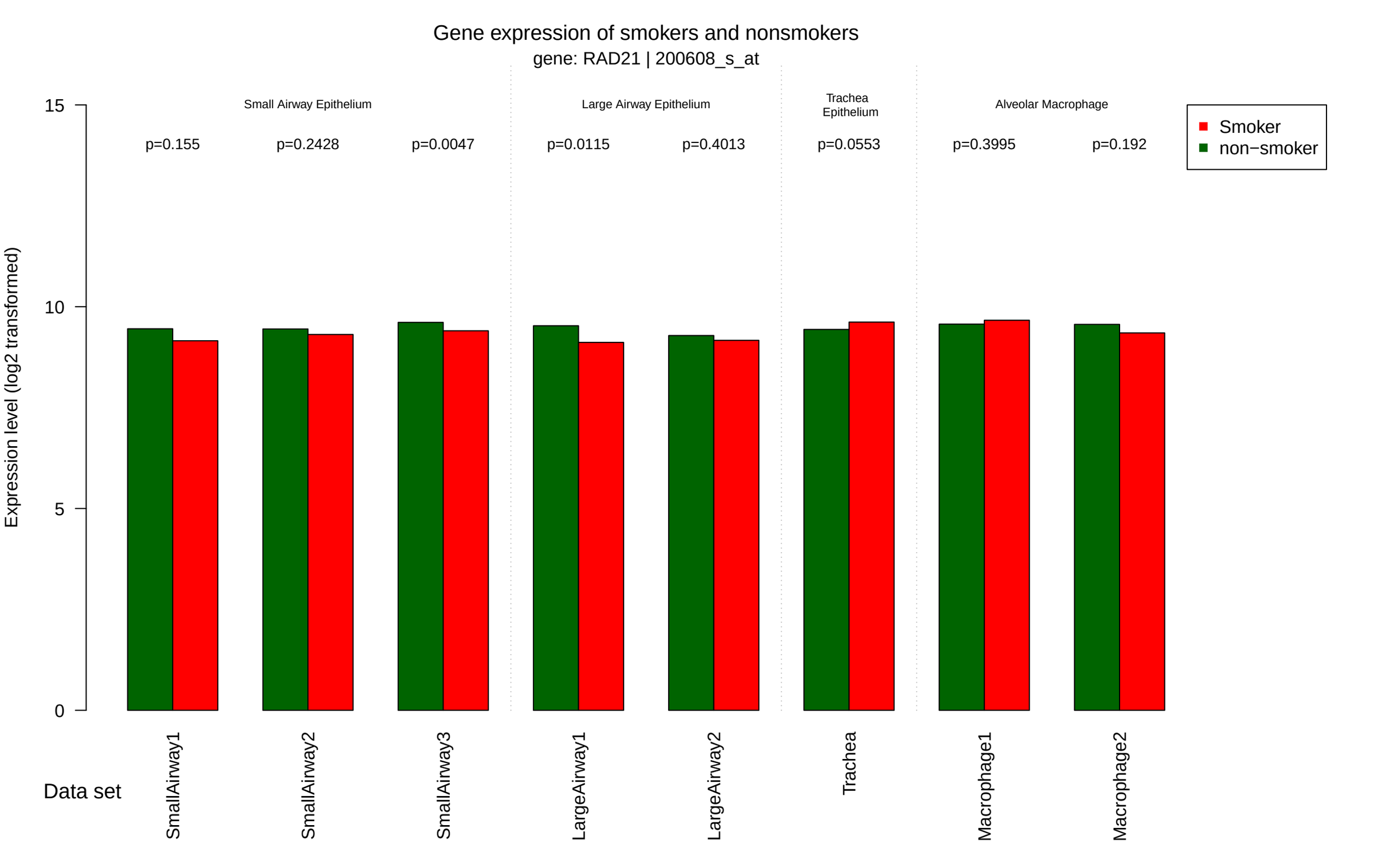
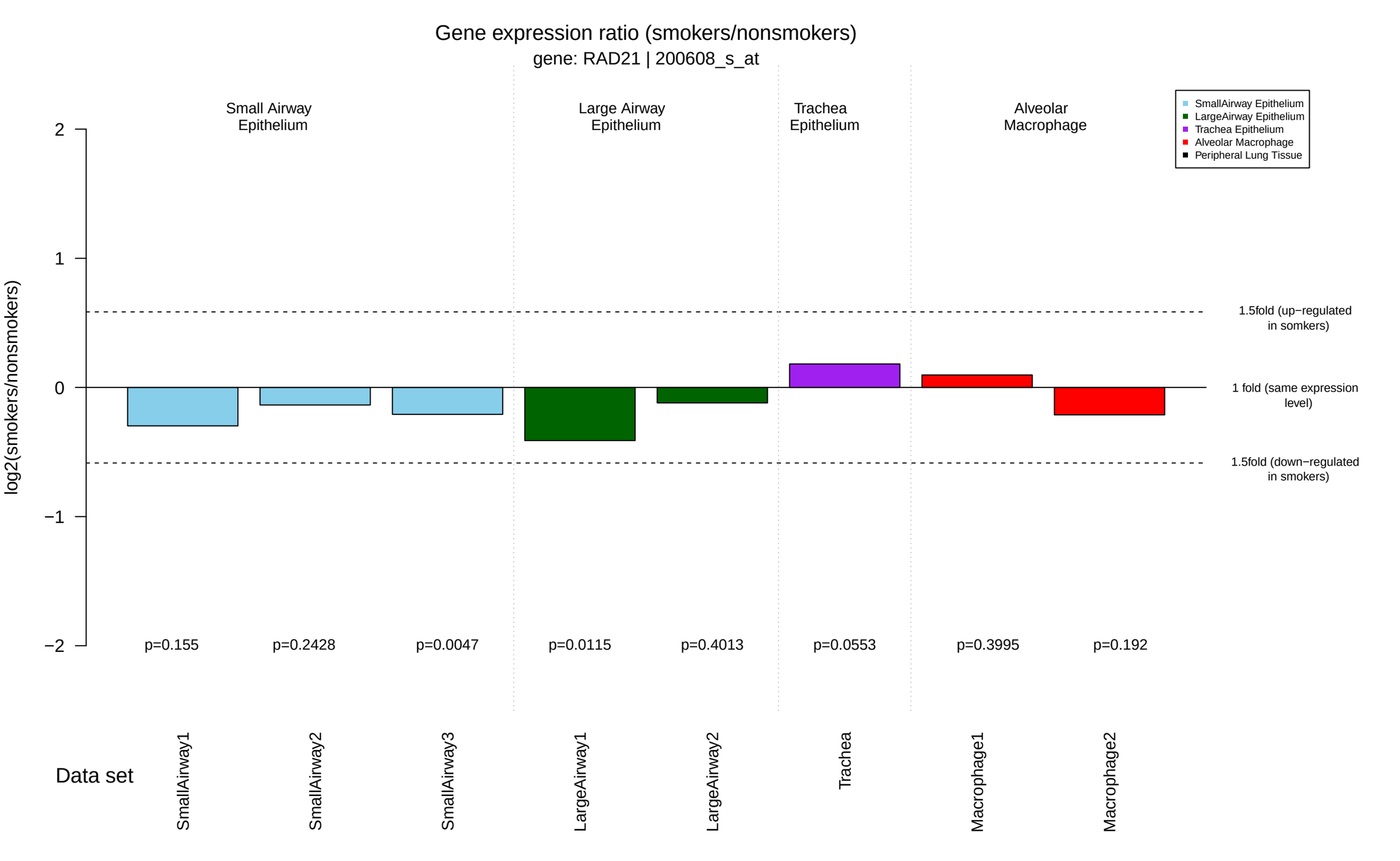
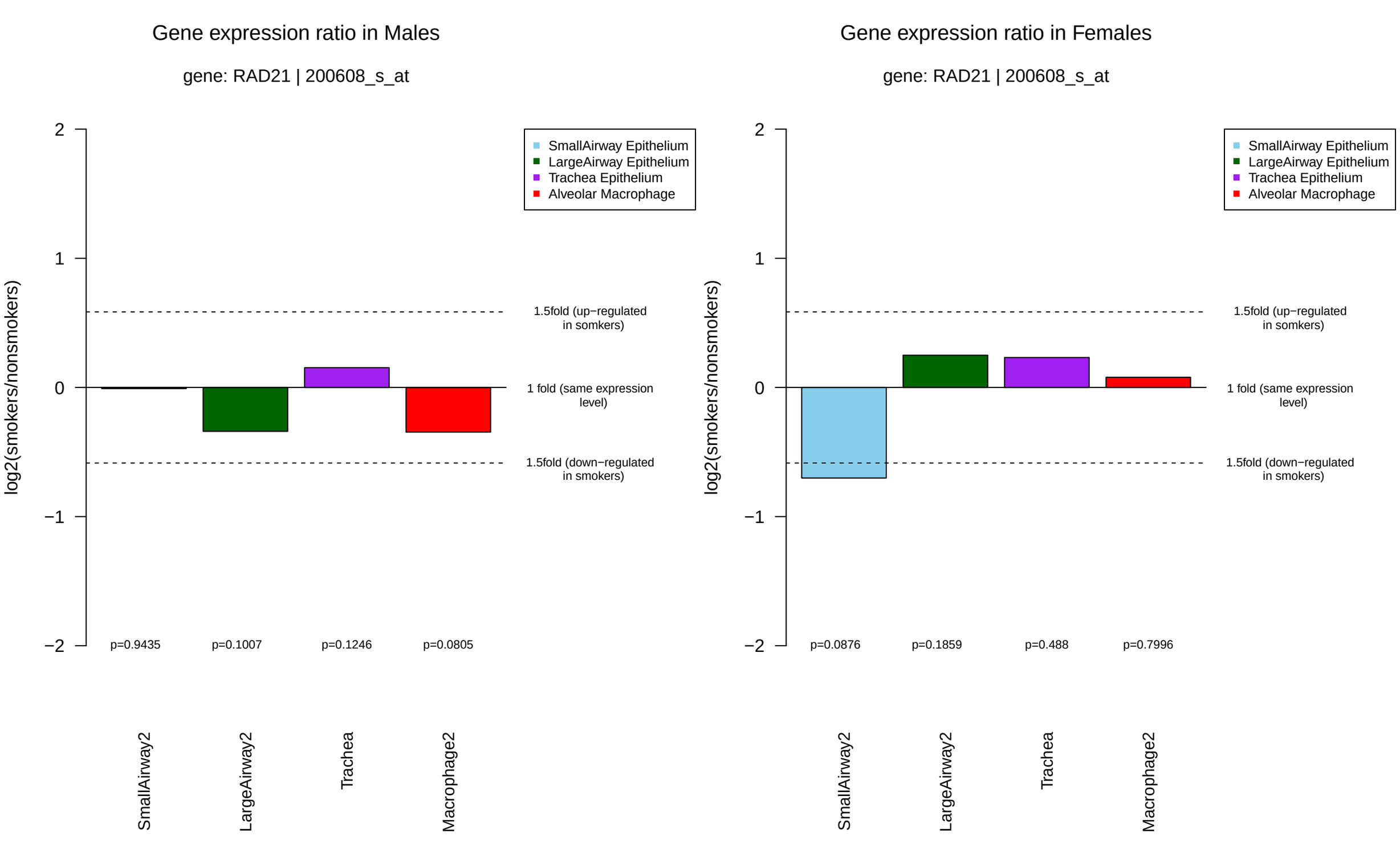
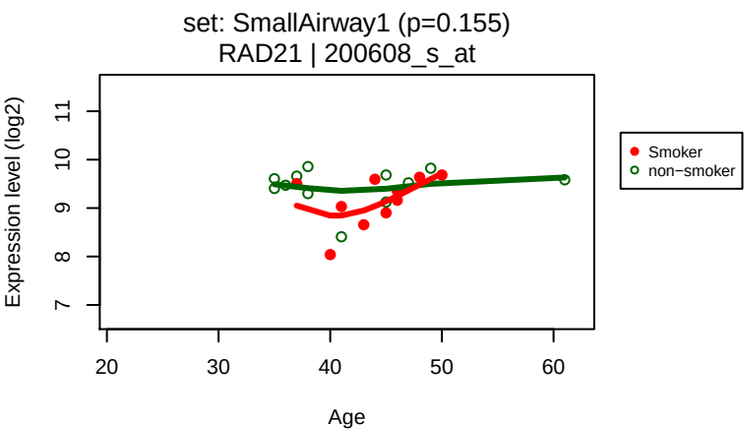
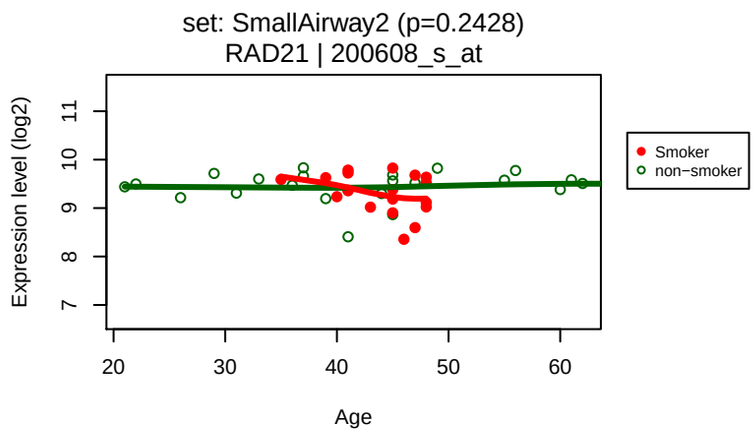
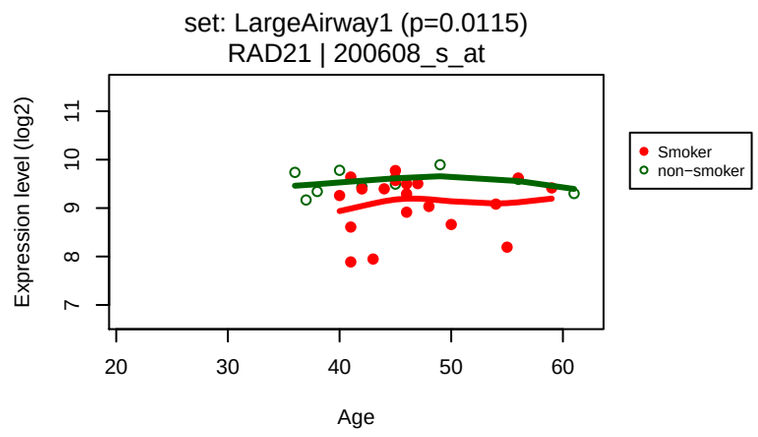

Gene expression of smokers and nonsmokers
gene: RAD21 | 200608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.155
p=0.2428
p=0.0047
p=0.0115
p=0.4013
p=0.0553
p=0.3995
p=0.192
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

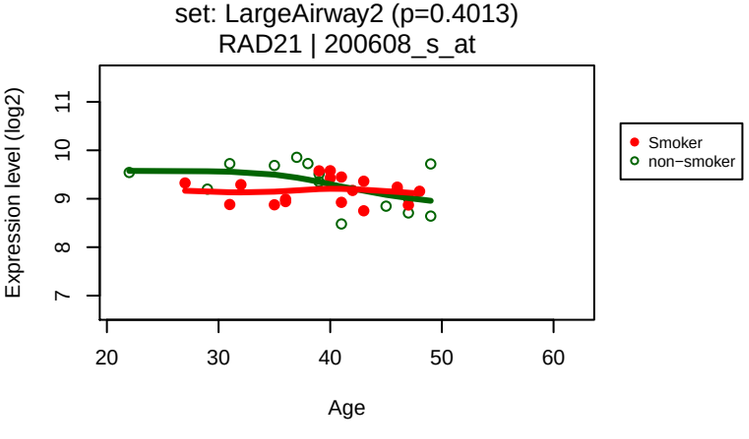

Gene expression ratio (smokers/nonsmokers)
gene: RAD21 | 200608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.155
p=0.2428
p=0.0047
p=0.0115
p=0.4013
p=0.0553
p=0.3995
p=0.192
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

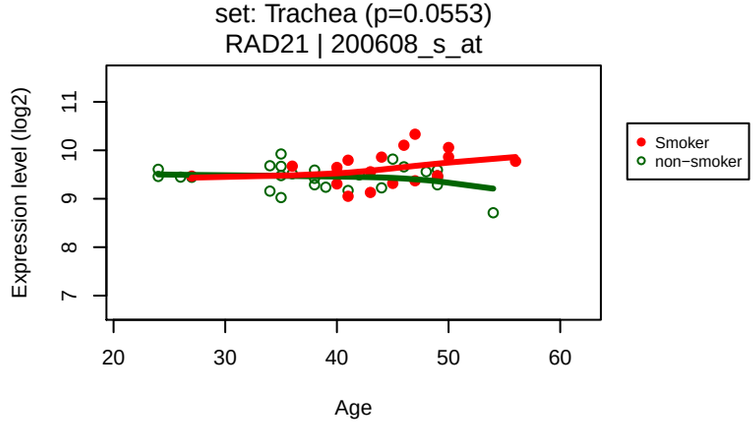

Gene expression ratio in Males
gene: RAD21 | 200608_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9435
p=0.1007
p=0.1246
p=0.0805
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD21 | 200608_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0876
p=0.1859
p=0.488
p=0.7996
SmallAirway2
LargeAirway2
Trachea
Macrophage2

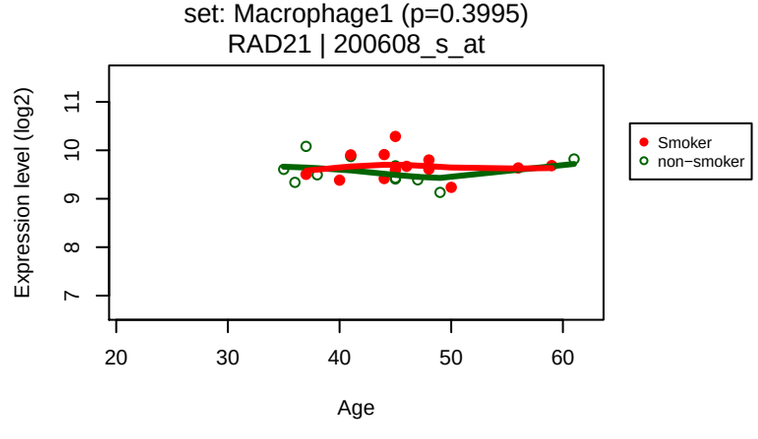

set: SmallAirway1 (p=0.155)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker

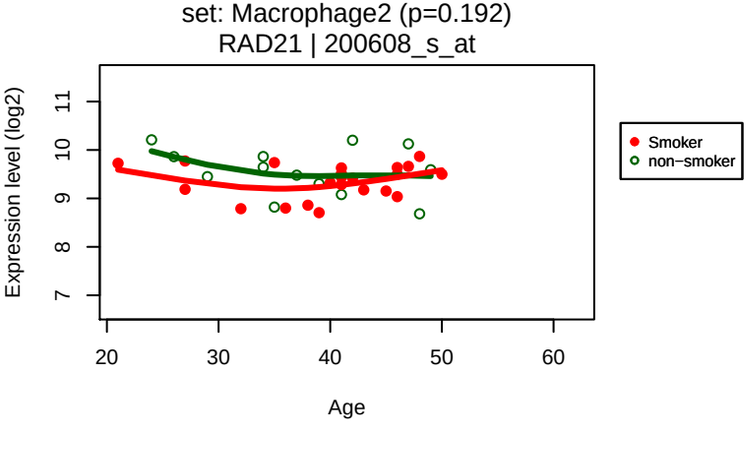

set: SmallAirway2 (p=0.2428)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker

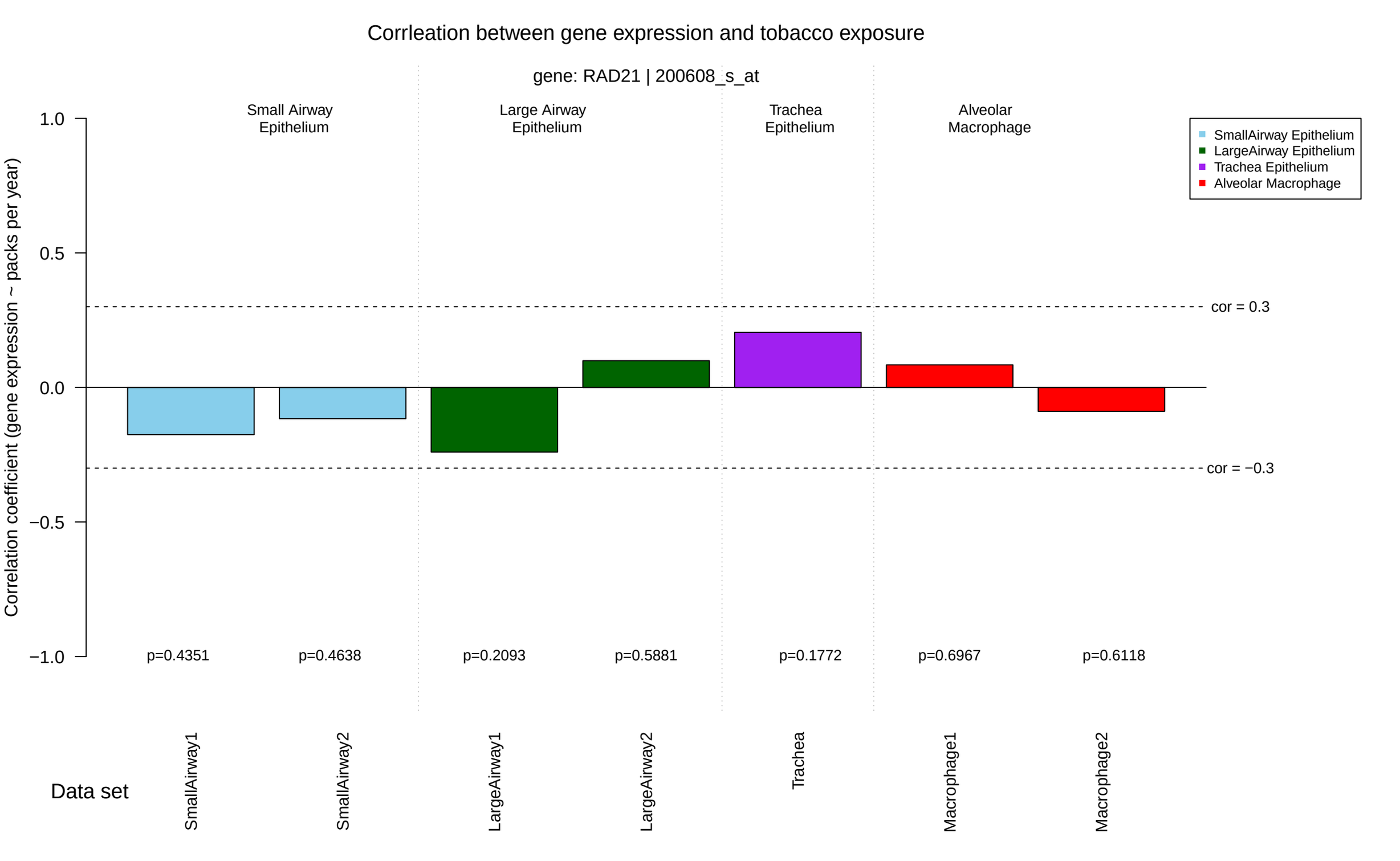

set: LargeAirway1 (p=0.0115)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4013)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0553)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3995)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.192)
RAD21 | 200608_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD21 | 200608_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4351
p=0.4638
p=0.2093
p=0.5881
p=0.1772
p=0.6967
p=0.6118
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2